TİMSAN
TRANSFORMATÖR SANAYİ
MALATYA

TRANSFORMER EXPERIMENT REPORT

AB-1440-T
T-20070
07.22

TÜRKAK
AB-1440-T

Customer							
kVA	1500 kVA	Voltage	11000 / 400 V.	# of Phase	3	Frequency	50 Hz.

Serial No	20070
Project No	2721009
Date	18.07.2022
Vector Gr	Dyn11
Cooling	ONAN

Pos	HV	LV	Primary Current	Secondary Current	HV Winding Resistances (TS_EN_60076_1 CLAUSE11.2) (T= 36 °C) R (A B)	R (B C)	R (C A)
1	10450		82,873				
2	10725		80,748				
3	11000	400 / √3	78,730	2165,064	0,9625	0,9638	0,9658
4	11275		76,809				
5	11550		74,981				
6							
7							

LV Winding Resistances (TS_EN_60076_1 CLAUSE11.2) (T= 36 °C)

R (a b)	0,001056	R (b c)	0,001048	R (c a)	0,001045

Pos	Nominal Voltage Ratio	Measured Values (No Load) (TS_EN_60076_1 CLAUSE11.3) AB / ab	BC / bc	CA / ca	Errors of Measured Values % (No Load) AB / ab	BC / bc	CA / ca
1	26,1250	26,14	26,14	26,14	0,057	0,057	0,057
2	26,8125	26,83	26,83	26,83	0,065	0,065	0,065
3	27,5000	27,49	27,49	27,49	0,036	0,036	0,036
4	28,1875	28,18	28,17	28,18	0,027	0,062	0,027
5	28,8750	28,87	28,87	28,86	0,017	0,017	0,052
6							
7							

Applied Voltage Test (50 Hz. 60 Sec.) (IEC 60076-3)				Induced Voltage Test (150 Hz. 40 Sec.) (IEC 60076-3)			
HV	28 kV	LV	3 kV	U = 800 V	I1 = A.	I2 = A.	I3 = A.

No Load Losses and Currents (TS_EN_60076_1 CLAUSE11.5)

Voltage Tr. Coeff. : 1	Current Tr. Coeff. : 1	Guaranteed Current (% Io): 1,1± %30	Guaranteed No Load Loss: 2500 W + % 0	
U1: 400 V	I1 : 8,06 A.	W1:	Measured Current (%)	Measured Loss (W)
U2: 400 V	I2 : 7,54 A.	W2:	0,350	1520
U3: 400 V	I3 : 7,13 A.	W3:		

Short Circuit and Load Loss Measurement (TS_EN_60076_1 CLAUSE11.4)

Guaranteed % Uk (75 °C) :			6 ± % 10		Guaranteed Load Loss (75 °)		22000 W + % 0	
Pos	Currnt.Tr.Coeff	Voltg.Tr.Coeff	Voltage	Current	WATT	Measured Pk	%Uk 75	Pk 75
3	1	1	665	78,730	18036	18036	6,07	20075

Isolation Resistances 5000 V. (20 °C) (TS_EN_60076_1 CLAUSE11.12)

t (sec)	0	15	30	45	60	P.E.	Power Factor (Tg δ) °C
HV-T (MΩ)			24000		36000	1,50	
HV-LV (MΩ)			18000		27000	1,50	
LV-T (MΩ)			30000		39000	1,30	

Oil			Load	Efficiency Cos α=1	Cos α=0,8	Voltage Regulation Cos α=1	Cos α=0,8
Type	Weight	Power fact(t= °C	4/4				
NYNAS Lyra X			3/4				
Oil Breakdown Voltage		kV / 2,5 mm	2/4				
74 72 72 70 71 70		Umean : 71,5 kV	1/4				

LEAKAGE TEST (TS_EN_60076_1 CLAUSE11.11) (0,30 Bar , 8 Hours)

Customer | Test Manager | Genr. Manager

TİMSAN
TRANSFORMATÖR SAN.TIC.LTD.ŞTİ
1.Organize Sanayi Bölgesi 5 Cad. MALATYA
Tel: (0422) 237 55 41-42-43 - Fax 237 55 44
Beydağı V.D: 844 022 2985

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TİMSAN TRANSFORMATÖR SANAYİ MALATYA	TİMSAN TEST LABORATORY 1.Organize sanayi bölgesi 44000 Yeşilyurt/Malatya/Turkey TEST REPORT	TÜRKAK AB-1440-T AB-1440-T T-20070 07.22

Customer name/address:		**Number of Experiment Period/Cycle:**	10
Report number:	T-20070	**Test result:**	Passed
Name and identity of test item:	1500 kVA EXPANSION TANK	**Date of Test:**	18.07.2022
Serial/Project Number:	20070 2721009	**Number of pages of the Report:**	3
The Date of Receipt of test item:	18.07.2022	**Test Stage:**	3
Remarks:	ONAN	**HV-LV Winding/Conductive Type:**	COPPER HW- LINE COND. LW-FOIL COND.

TİMSAN TEST LABORATORY accredited by TÜRKAK under registration number [AB-1440-T accreditation file no] for TS EN ISO/IEC 17025 as test laboratory"

Turkish Accreditation Agency (TURKAK) is a signatory to the European co-operation for Accreditation (EA) Multilateral Agreement (MLA) and to the International Laboratory Accreditation Cooperation (ILAC) Mutual Recognition Arrangement (MRA) for the recognition of test reports

The test and/or measurement results, the uncertainties (if applicable) with confidence probability and test methods are given on the following pages which are part of this report

Seal	Experiment Start-End Date	Tested Manager	General Manager
TİMSAN TRANSFORMATOR SAN.TIC.LTD.ŞTİ. 1.Organize Sanayi Bölgesi 5.Cad. MALATYA Tel: (0422) 237 68 41-42-43 - Fax:237 68 44 Beydağı V.D. 844 022 2981	18.07.2022	Electrical Electronics Engineer TİMSAN TRANSFORMATOR SAN.TIC.LTD.ŞTİ. 1.Organize Sanayi Bölgesi 5.Cad. MALATYA Tel: (0422) 237 68 41-42-43 - Fax:237 68 44 Beydağı V.D. 844 022 2981	TİMSA TRANSFORMATOR SAN. 1.Organize Sanayi Bölgesi 5. Tel: (0422) 237 68 41-42-43 Beydağı V.D. 844 0

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